

Surfaces and Area

Lecture 07

Instructor: Asgher Ali

October 9th, 2025

Recap

We have defined curves in the plane in three different ways:

Form	Mathematical Representation
Explicit	$y = f(x)$
Implicit	$F(x, y) = 0$
Parametric vector	$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j}, \quad a \leq t \leq b$

We have analogous definitions of surfaces in space:

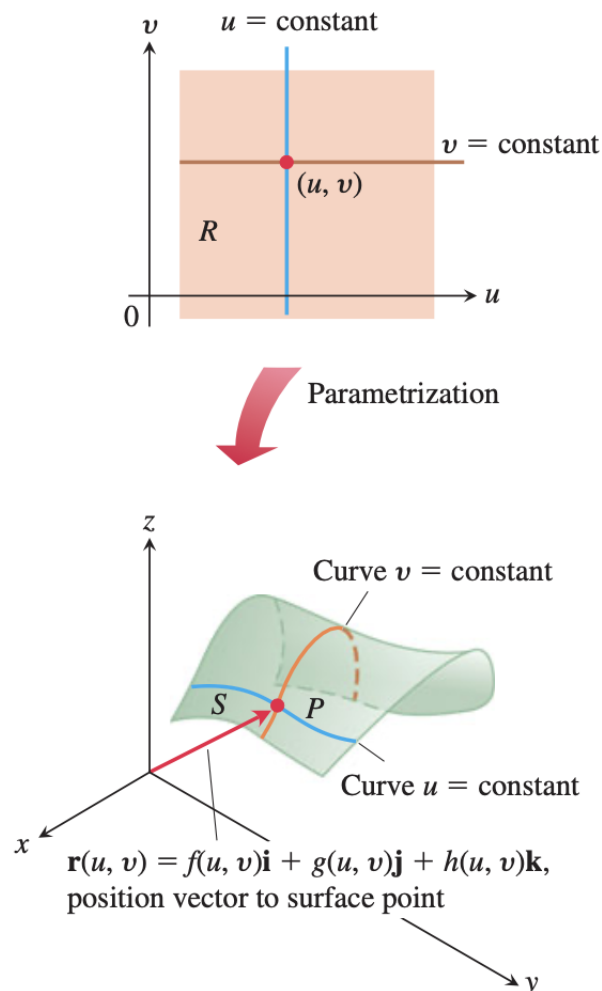
Form	Mathematical Representation
Explicit	$z = f(x, y)$
Implicit	$F(x, y, z) = 0$

Parametrization of Surfaces

Suppose that

$$\mathbf{r}(u, v) = f(u, v)\mathbf{i} + g(u, v)\mathbf{j} + h(u, v)\mathbf{k}$$

and



1. $\mathbf{r}$ is a continuous function that is defined in the uv -plane and one-to-one on the interior of R .
2. We call the **range** of $\mathbf{r}$ the SURFACE S defined or traced by $\mathbf{r}$.
3. The equation above with the domain R constitutes a **parametrization of the surface**.
4. The variable u and v are the **parameters** and R is the **parametric domain**.

5. To simplify our discussion, we take R to be a rectangle defined by inequalities of the form $a \leq u \leq b, c \leq v \leq d$.
6. The requirement that $\mathbf{r}$ to be one-to-one on the interior of R ensures that S does not cross itself.
7. The above equation is the vector equivalent of three parametric equations:

$$x = f(u, v)$$

$$y = g(u, v)$$

$$z = h(u, v)$$

Examples

1. Find the parametrization of the cone

$$z = \sqrt{x^2 + y^2}, \quad 0 \leq z \leq 1$$

2. Find a parametrization of the sphere $x^2 + y^2 + z^2 = a^2$.
3. Find the parametrization of the cylinder

$$x^2 + (y - 3)^2 = 9, \quad 0 \leq z \leq 5$$

-
1. Here we can use cylindrical coordinates to get the right parametrization. A typical point (x, y, z) on the cone has

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = \sqrt{x^2 + y^2} = r$$

with $0 \leq r \leq 1$ and $0 \leq \theta \leq 2\pi$. Take $u = r$ and $v = \theta$, and

$$\mathbf{r}(u, v) = u \cos v \mathbf{i} + u \sin v \mathbf{j} + u \mathbf{k}$$

with $0 \leq u \leq 1$ and $0 \leq v \leq 2\pi$. The parametrization is one-to-one on the interior of the domain R , though not on the boundary tip of its cone where $r = 0$.

2. Here we can use spherical coordinates to get the right parametrization. A typical point (x, y, z) on the sphere has

$$x = a \sin \phi \cos \theta$$

$$y = a \sin \phi \sin \theta$$

$$z = a \cos \phi$$

with $0 \leq \phi \leq \pi$ and $0 \leq \theta \leq 2\pi$. Take $u = \phi$ and $v = \theta$, and

$$\mathbf{r}(u, v) = a \sin u \cos v \mathbf{i} + a \sin u \sin v \mathbf{j} + a \cos u \mathbf{k}$$

with $0 \leq u \leq \pi$ and $0 \leq v \leq 2\pi$. The parametrization is one-to-one on the interior of the domain R , though not on the boundary boundary poles where $\phi = 0$ or $\phi = \pi$.

3. **Hint:** $r = 6 \sin \theta$, $0 \leq \theta \leq \pi$ and

$$\mathbf{r}(u, v) = 3 \sin 2u \mathbf{i} + 6 \sin^2 u \mathbf{j} + z \mathbf{k}$$

with $0 \leq u \leq \pi$ and $0 \leq z \leq 5$.

Smooth Surface

A parametrized surface $\mathbf{r}(u, v) = f(u, v) \mathbf{i} + g(u, v) \mathbf{j} + h(u, v) \mathbf{k}$ is **smooth** if $\mathbf{r}_u$ and $\mathbf{r}_v$ are continuous and $\mathbf{r}_u \times \mathbf{r}_v$ is never zero on the interior of the parameter domain.

Area of the smooth surface

The AREA OF THE SMOOTH SURFACE

$$\mathbf{r}(u, v) = f(u, v) \mathbf{i} + g(u, v) \mathbf{j} + h(u, v) \mathbf{k}$$

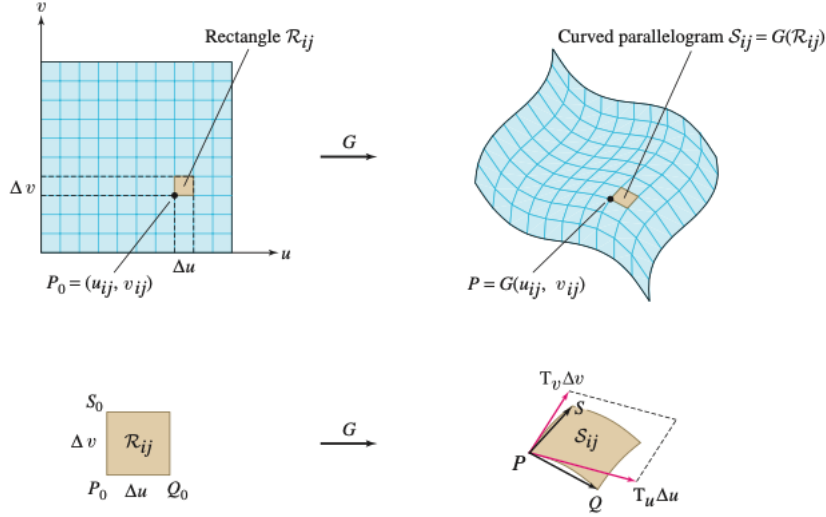
with $a \leq u \leq b$, $c \leq v \leq d$ is

$$\begin{aligned} A &= \iint_{\mathcal{D}} |\mathbf{r}_u \times \mathbf{r}_v| dA \\ &= \int_c^d \int_a^b |\mathbf{r}_u \times \mathbf{r}_v| du dv \end{aligned}$$

or equivalently

$$\begin{aligned} A &= \iint_{\mathcal{D}} |\mathbf{T}_u \times \mathbf{T}_v| dA \\ &= \int_c^d \int_a^b |\mathbf{T}_u \times \mathbf{T}_v| du dv \end{aligned}$$

- Assume for simplicity that $\mathcal{D}$ is a rectangle (the argument also applies to general domains).
- Divide $\mathcal{D}$ into a grid of small rectangles $\mathcal{R}_{ij}$ of sizes $\Delta u \times \Delta v$.
- Compare the area of $\mathcal{R}_{ij}$ with the area of its image under G .
- This image is a "curved" parallelogram $\mathcal{S}_{ij} = G(\mathcal{R}_{ij})$



- If Δu and Δv are small, then the curved parallelogram $\mathcal{S}_{ij}$ has approximately the same area as the "genuine" parallelogram with sides $\overrightarrow{PQ}$ and $\overrightarrow{PS}$.

This can be written as

$$\text{Area}(\mathcal{S}_{ij}) \approx |\overrightarrow{PQ} \times \overrightarrow{PS}|$$

where

$$\begin{aligned} \overrightarrow{PQ} &= G(u_{ij} + \Delta u, v_{ij}) - G(u_{ij}, v_{ij}) \\ &\approx \frac{\partial G}{\partial u} \Delta u = \mathbf{T}_u \Delta u \end{aligned}$$

and

$$\begin{aligned} \overrightarrow{PS} &= G(u_{ij}, v_{ij} + \Delta v) - G(u_{ij}, v_{ij}) \\ &\approx \frac{\partial G}{\partial v} \Delta v = \mathbf{T}_v \Delta v \end{aligned}$$

This implies that

$$\begin{aligned} \text{Area}(\mathcal{S}_{ij}) &\approx |\mathbf{T}_u \Delta u \times \mathbf{T}_v \Delta v| \\ &= |\mathbf{T}_u \times \mathbf{T}_v| \Delta u \Delta v \end{aligned}$$

To compute the surface area of S , we assume that G is one-to-one, except possibly on the boundary of $\mathcal{D}$. We also assume that G

is regular, except possibly on the boundary of $\mathcal{D}$. Recall that "regular means" $\mathbf{T}_u \times \mathbf{T}_v$ or $\mathbf{r}_u \times \mathbf{r}_v$ is non-zero.

The entire surface $\mathcal{S}$ is made of the small patches $\mathcal{S}_{ij}$, so we can apply approximation on each patch to obtain

$$\text{Area}(\mathcal{S}) = \sum_{ij} \text{Area}(\mathcal{S}_{ij})$$

The sum is Riemann sum and when Δu and Δv tend to zero, the Riemann sum converges to the double integral given below

$$\text{Area}(\mathcal{S}) = \iint_{\mathcal{D}} |\mathbf{r}_u \times \mathbf{r}_v| dA$$

Surface area Differential for a Parametrized Surface

$$d\sigma = |\mathbf{r}_u \times \mathbf{r}_v| du dv$$

Surface Area Differential

$$\iint_{\mathcal{S}} d\sigma$$

Differential formula for surface area

Examples

1. Find the surface area of the cone $z = \sqrt{x^2 + y^2}, 0 \leq z \leq 1$
2. Find the surface area of a sphere with radius a .
3. Let S be the "football" surface formed by rotating the curve $x = \cos z, y = 0, -\frac{\pi}{2} \leq z \leq \frac{\pi}{2}$ around the z -axis. Find a parametrization for S and compute its surface area.

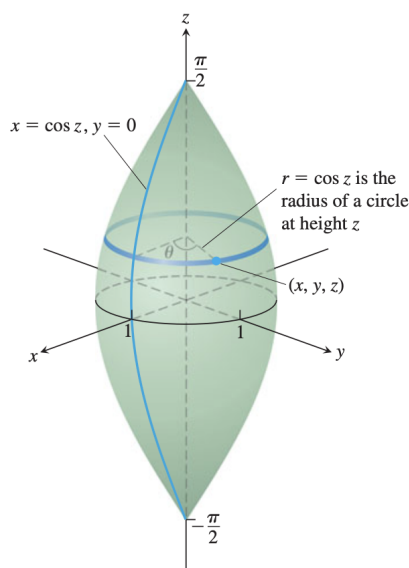


Figure 1: The football surface obtained by rotating the curve $x = \cos z$ about the z -axis.